Unit 3 (Part 3) – Working with Rationals

Proving a Real Number is Rational

Rational numbers are the set of all real numbers that can be expressed as a quotient of integers:

$$\mathbb{Q} = \left\{ q \mid \ q = \frac{a}{b} \ ; \ a, b \in \mathbb{Z} \ ; \ b \ \neq 0 \right\}$$

To prove that a real number is rational one needs to express the value as a quotient of two integers (in simplest form).

A recurring decimal is a decimal that continues et infinitum. For each number that recurs, a solid dot is placed above the number (or dots). As recurring decimals are rational numbers, they can be converted into fractions.

Examples:

i. Convert $2,\dot{5}$ into a fraction

ii. Convert $-0,\dot{4}\dot{8}$ into a fraction.

iii. Convert $2,5\overline{73}$ into a fraction

Finding rational numbers between two values

Example 1:

Find a rational number between $\frac{5}{7}$ and $\frac{3}{4}$.

Example 2:

Between which two consecutive integers do the following irrational numbers lie?

(a) $\sqrt{12}$

(b) $\sqrt[3]{20}$

(c) -2π

Homework: Ex. 2 pg. 10 (Mind Action Series new edition, 2022)